40018 43

PMT 3: Relations

Submitters

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kw1425

Emarking

9) (i) FALSE. for

$A = \{1, 2, 3\}$

$R = \{\langle 1, 2\rangle\}$ $R \subseteq A^2$

$R^{-1} = \{\langle 2, 1\rangle\}$

$R \circ R^{-1} = \{\langle 1,1\rangle, \langle 2,2\rangle\}$

↳ missing $\langle 3,3\rangle$

(ii) Reflexive ⇒ symmetric

FALSE. for

$A = \{1, 2, 3\}$ $R \subseteq A^2$

$R = \{\langle 1,1\rangle, \langle 2,2\rangle, \langle 3,3\rangle, \langle 1,2\rangle\}$

↳ reflexive as $Id_A \subseteq R$.

NOT symmetric as $\langle 2,1\rangle$ is missing

(iii) claim symmetric ⇒ Reflexive

FALSE for

$A = \{1, 2, 3\}$ $R \subseteq A^2$

$R = \{\langle 1,2\rangle, \langle 2,1\rangle\}$

↳ symmetric as $R = R^{-1}$

↳ NOT reflexive as does not contain Id_A.

~~8)(i) Proof~~

9) (iv) ~~proof~~ Assume R, S on A are symmetric

yes!

take a, b ∈ A, p ∈ R∪S

then, $\langle a,b\rangle \in R\cup S$, $a R\cup S\, b$.

we need to show $a R\cup S\, b \Rightarrow b R\cup S\, a$.

by definition of ∪, aRb or aSb

by cases...

1. Assume aRb. Since R is symmetric, '$aRb \Rightarrow bRa$'.

first, showing 'or' then use ∪.

By definition of ∪, $b R\cup S\, a$

2. Assume aSb. Similarly, '$aSb \Rightarrow bSa$', so $b R\cup S\, a$

So, $b R\cup S\, a$ holds with no assumptions.

So, 'R and S are symmetric implies R∪S is symmetric'

9) (v) <u>claim</u> take transitive relation R; $\overline{R}$ is transitive

FALSE for

$A = \{1,2,3\}$ $R \subseteq A^2$

$R = \{\langle 1,1\rangle, \langle 2,2\rangle, \langle 3,3\rangle\}$

$\overline{R} = \{\langle 1,2\rangle, \langle 1,3\rangle, \langle 2,1\rangle, \langle 2,3\rangle,$
$\langle 3,1\rangle, \langle 3,2\rangle\}$

↳ not transitive, as eg $\langle 1,2\rangle, \langle 2,1\rangle \in \overline{R}$ but $\langle 1,1\rangle \notin \overline{R}$. Yes!

(vi) <u>claim</u> take a non symmetric relation R; $\overline{R}$ is symmetric.

FALSE for

$A = \{1,2\}$ $R \subseteq A^2$

$R = \{\langle 1,2\rangle\}$

$\overline{R} = \{\langle 1,1\rangle, \langle 2,1\rangle\ \langle 2,2\rangle\}$

↳ not symmetric

10) (i) $A = \{1,2,3\cancel{\}},4\}$

$R = \{\langle 1,2\rangle\ \langle 2,3\rangle\ \langle 1,3\rangle\}$

$S = \{\langle 2,2\rangle\ \langle 2,4\rangle\}$

$R \cup S = \{\langle 1,2\rangle\ \langle 2,3\rangle\ \langle 1,3\rangle\ \langle 2,2\rangle$
$\langle 2,4\rangle\}$

↳ not transitive as missing $\langle 1,4\rangle$

10)(ii) claim if R, S are transitive, $R \cap S$ is transitive.

proof Assumme R, S are transitive

~~take $a, b \in A$ ... $p \in$ ... $R \cap S$.~~

~~then $\langle a, b \rangle \in R \cap S$, then $a R\cap S b$~~

~~by definition of $\cap$, aRb and aSb.~~

take $a, b, c, \in A$ State the LHS of transitive def here

suppose $a R\cap S b$ and $b R\cap S c$.

we must show that $a R\cap S c$.

by definition of $\cap$, aRb, bRc and aSb, bSc.

more specific!

Since they are transitive by assumption, aRc and aSc, so $a R\cap S c$. Yes!

so, $a R\cap S b$ and $b R\cap S c \Rightarrow a R\cap S c$

so R, S are transitive implies $R \cap S$ is transitive.

converse not true for

$A = \{1, 2, 3\}$ $R \subseteq A^2$

$R = \{\langle 1,1 \rangle, \langle 1,2 \rangle, \langle 2,3 \rangle\} \rightarrow$ not transitive

$S = \{\langle 1,1 \rangle \langle 1,2 \rangle\} \rightarrow$ transitive

$R \cap S = \{\langle 1,1 \rangle \langle 1,2 \rangle\} \rightarrow$ transitive

$R \cap S$ being transitive does not imply R and S are transitive.

Yes!

10) (iii) $A = \{1, 2, 3, 4\}$ $R \subseteq A^2$

$R = \{\langle 1, 2\rangle \langle 3, 4\rangle\}$ transitive

$S = \{\langle 2, 3\rangle \langle 4, 1\rangle\}$

$R \circ S = \{\langle 1, 3\rangle, \langle 3, 1\rangle\}$

↳ missing $\langle 1, 1\rangle$, $\langle 3, 3\rangle$

Yes!

ii) <u>claim</u> R is an equivelance relation on A iff R is reflexive and circular on A.

<u>proof</u>.

1- "If R is an equivalence relation on A, R is reflexive and circular on A."

Assume R is an equivalence relation on A.
then by definition R is reflexive ~~on A~~, symmetric and transitive on A.
so, for all $a, c \in A$, if there exists a b such that aRb and bRc, then aRc by definition of transitivity.
then, $aRc \Rightarrow cRa$ by definition of symmetry.
so, $\forall a, b \in A\ (\exists b(aRb \wedge bRc) \Rightarrow cRa)$
so, R is circular, ~~ref~~ and reflexive if it is an equivalence relation, as required.

2. if R is reflexive and circular, R is an equivalence relation on A!

Assume R is reflexive and circular.
We need to show that R is reflexive, symmetric and transitive.
1. ~~We~~ Reflexive: true by assumption.
2. Symmetric: take $x, y \in A$ ~~an~~ and xRx and xRy. We know xRx must hold, as R is reflexive and $xRx \wedge xRy \Rightarrow yRx$ by definition of circular. For the formula to

hold $xRy \Rightarrow yRx$. So, R is symmetric.

3. Transitive: since $xRy \wedge yRz \Rightarrow zRx$ has circularity?

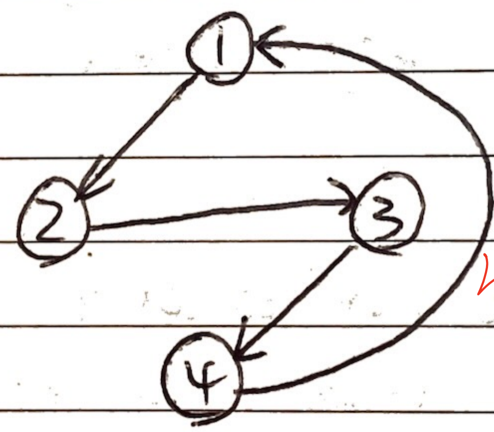
$\Rightarrow xRz$ by symmetry, R is transitive.

Therefore R is symmetric, transitive and reflexive, so R is an equivalence relation on A.

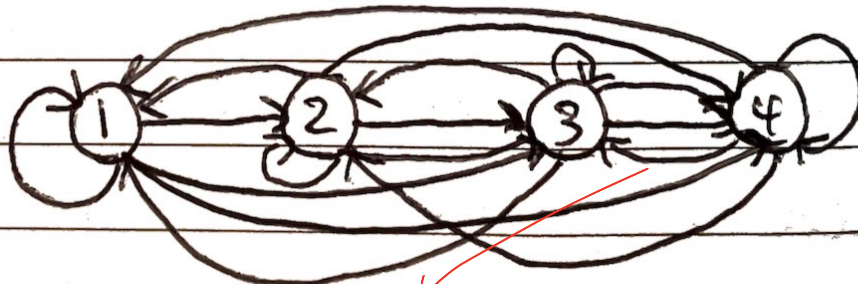
12) (i) $R^+ = \{\langle 1,2\rangle, \langle 2,3\rangle, \langle 3,4\rangle, \langle 4,1\rangle,$
$\langle 1,3\rangle, \langle 1,4\rangle, \langle 1,1\rangle, \langle 2,4\rangle,$
$\langle 2,1\rangle, \langle 2,2\rangle, \langle 3,1\rangle, \langle 3,2\rangle,$
$\langle 3,3\rangle, \langle 4,2\rangle, \langle 4,3\rangle, \langle 4,4\rangle\}$

Yes!

R:



R^+:



1 2

3 4

gives a clearer diagram

(ii) $R = \{\langle a,b\rangle \in \mathbb{N}^2 \mid b = 2x\}$

unfortunately, this was a typo in question!

$b = 2a$

$= \{\langle 1,2\rangle\ \langle 1,4\rangle\ \langle 1,6\rangle \ldots$
$\langle 2,2\rangle\ \langle 2,4\rangle\ \langle 2,6\rangle \ldots$
$\langle 3,2\rangle\ \langle 3,4\rangle\ \langle 3,6\rangle \ldots$
$\ldots$
$\}$

$R^+ = R$

$= \{ \langle a, b \rangle \mid b = 2x3$

$aR^+b \triangleq \exists n (b = 2^n a)$

13) $A = \{1, 2, 3\}$ $\quad R \subseteq A^2$

$R = \{\langle 1,2 \rangle, \langle 2,3 \rangle \langle 3,1 \rangle\}$

$R^2 = \{\langle 1,3 \rangle \langle 2,1 \rangle \langle 3,2 \rangle\}$

$|A| = 3.$

$R \cup R^2 = \{\langle 1,2 \rangle \langle 1,3 \rangle \langle 2,1 \rangle \langle 2,3 \rangle$
$\langle 3,1 \rangle \langle 3,2 \rangle\}$

$R^+ = \{\langle 1,1 \rangle \langle 1,2 \rangle \langle 1,3 \rangle$
$\langle 2,1 \rangle \langle 2,2 \rangle \langle 2,3 \rangle$
$\langle 3,1 \rangle \langle 3,2 \rangle \langle 3,3 \rangle.$

$\Rightarrow R^+ \neq R \cup R^2$

Yes!